

Computational and Experimental Mechanistic Insights into the Ethanol-to-Butanol Upgrading Reaction over MgO

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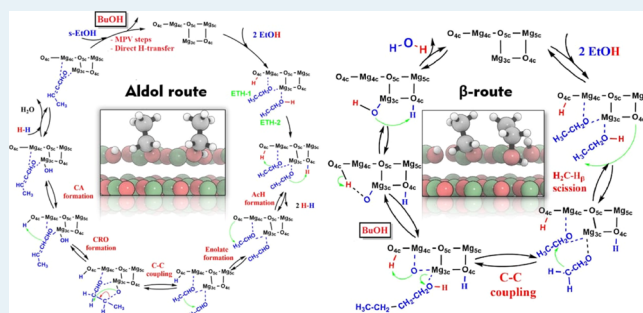


Supporting Information

ABSTRACT: The mechanism of ethanol upgrading to higher products is still under debate, especially regarding intermediate species and hydrogenation and dehydrogenation steps. In this work, we conducted a combined theoretical and experimental approach to contribute to this discussion. For such, detailed electronic structure density functional theory calculations (aiming at probing density of states, infrared spectra, geometric parameters, charge densities, and reaction energetics) and diffuse reflectance infrared Fourier transform spectroscopy experiments were carried out revealing the relevance of an appropriate combination of reactive surface sites to support the formation of several intermediates that are formed in the C–C coupling over MgO.

The roles of Mg and O sites were also studied under an electronic perspective and different geometrical arrangements. We found that a kink configuration was the most adequate for ethanol to 1-butanol upgrading. Our calculations also gave us arguments to propose distinct reaction routes, whose mutual predominance would depend upon reaction temperature. At temperatures up to 573 K, the so-called β -route, which goes through scission of a C_{β} –H bond and formation of an oxametallacycle-like intermediate, would dominate the coupling, whereas at higher temperatures, up to 673 K, a more usual Guerbet mechanism, via an aldol coupling step and then consecutive hydrogenations, would be expected. The theoretical conclusions were followed by a careful experimental strategy using sequential experimental planning techniques in order to estimate accurate parameters with the lowest possible experimental load. Information from these different sources was coupled to develop a mathematical model for the rate of the ethanol upgrading reaction, using a Langmuir–Hinshelwood–Hougen–Watson approach. The developed and statistically validated model adequately described the experimental data at 673 K and 1.1 bar total pressure for ethanol partial pressures in the range from 0 to 20 kPa.

KEYWORDS: ethanol upgrading, DFT, 1-butanol, reaction mechanism, MgO catalyst, experimental planning



1. INTRODUCTION

The ever-growing energy demands associated with increasing environmental constraints make ethanol (EtOH) a very attractive energy source.¹ Such a fact emerges from a combination of lower emissions and toxicity with high energy density and, most importantly, because EtOH can be relatively easily obtained from different types of renewable sources (biomass), especially corn (USA) and sugarcane (Brazil) as well as from many other starch-rich crops like wheat and barley.² EtOH is also known as a vital platform molecule, used as a building block for commercially relevant products in the chemical industry.³ It can be easily transported, is sulfur-free, and can be converted into gaseous hydrogen with steam and autothermal reforming.⁴ Moreover, EtOH can be used directly as a net zero-carbon emission fuel (or fuel blend) and upgraded into more valuable chemicals with applications of enormous importance to the chemical industry, including (but not limited to) ethylene, acetone, ethyl acetate, diethyl ether, 1,3-butadiene (BD), and 1-butanol (BuOH).⁵

In particular, BuOH is receiving significant attention in the last few decades.^{6–9} Besides being vital to the production of acrylic and acid esters and largely used as a solvent, BuOH presents some advantages when compared to EtOH as a fuel: (i) larger energy density (~90% that of gasoline); (ii) less hygroscopicity and corrosivity; and (iii) larger miscibility in diesel and in gasoline. Moreover, BuOH can be used in regular combustion engines with only a few or even no further adjustments.¹⁰ On the other hand, the process of upgrading EtOH to BuOH is still economically unattractive^{11–14} and, therefore, attempts to develop alternative EtOH-upgrading routes and contributions to unraveling the role played by

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catalytic surfaces are needed in order to increase the efficiency of the known processes. To this end, the nature of catalytic surfaces must be further understood, along with well-engineered chemical reaction conditions because competitive reactions might occur affecting the desired product yields. For example, side-reactions like methane formation and dehydration into olefins and ethers may represent classes of unwanted byproducts in the pathway to valuable longer-chain products.¹⁵

Over the years, multicomponent, metal-supported, doped, and pure metal oxide-based materials have been tested as catalysts in EtOH-upgrading processes.^{14,16–20} Among the most studied oxide surfaces, MgO-based materials receive special attention because of their interesting catalytic performance, which is usually related to their surface acidic and basic properties, in turn pivotal in C–C coupling reactions.^{9,15,21,22} Although some researchers attempted to improve catalytic performance by doping MgO with metals (Sn, Pb, Zn, Cu, Ce, Zr, Ca, and Ba), no substantial improvements have been noticed in the selectivity likely because the presence of these metals altered the acid-basic site properties by decreasing the amount of strong basic sites necessary for the activation of EtOH.¹² In a different approach, MgO could be mixed with a second metal, preferably a stronger Lewis acid compared to Mg, as, for example, Al^{23,24} and Zr.²⁵ The main conclusion is that the rate of unwanted EtOH dehydration increased around one order of magnitude because of the higher density of stronger Lewis acid sites.

Regarding the interaction and chemical reactions of EtOH over MgO-based catalysts, the Guerbet route is usually accepted as a preferential pathway for the coupling of EtOH to BuOH.^{20,21,26–28} Accordingly, EtOH is first dehydrogenated to form acetaldehyde (AcH), following a self-aldolization reaction to generate 3-alkoxybutanal species; then, a dehydration reaction occurs to form crotonaldehyde (CRO) and finally BuOH is formed through sequential hydrogenation of the unsaturated carbon chain. However, side reactions forming numerous intermediates as well as undesired byproducts are competing for surface-active sites and for that reason not all steps may be fully explained by Guerbet chemistry. An alternative mechanism has been proposed and, overall, it strongly disagrees with the Guerbet community by assuming that EtOH molecules may combine directly via C–C coupling with negligible participation of AcH as an intermediate.^{14,16,29}

Nevertheless, both direct and indirect (Guerbet) communities do not always provide such strong evidence as to the mechanistic details and, therefore, they are unable to address all features that may be relevant in one or another aspect of the coupling process. Thus, conclusions concerning the role played by and the nature of surface (acid–base) active sites both on Mg as well as on many other metal-based oxides are still heavily debated, even though extensively studied.^{6,14,21,22,30,31} In fact, the underlying mechanisms that determine the distribution of intermediates and products are not well understood and need to be further revealed.

Given these motivating issues, we found it essential to investigate deeper how MgO surfaces influence the key steps of EtOH-upgrading reactions. As a direct consequence, the design of more stable, active, and selective basic metal oxide-based catalysts might be benefited by theoretical investigations into the mechanistic aspect. To tackle these issues, this work provides, by means of thorough density functional theory

(DFT) calculations, results supporting that the Guerbet-like aldol condensation mechanism,^{7,8,20,25,32} instead of a C β -activated direct coupling,^{14,16,22,33} represents the preferred pathway to the EtOH-to-BuOH upgrading process at temperatures higher than 573 K. For that, detailed electronic structure calculations and diffuse reflectance infrared Fourier transform spectroscopy (DRIFTS) experiments were carried out revealing the most significant aspects of the active center and reaction intermediates. Furthermore, a careful experimental strategy, based upon the sequential design of experiments, is presented to conduct kinetic studies in order to validate the computational results.

2. METHODOLOGY

2.1. Computational Procedure. DFT calculations were carried out using the supercell pseudopotential method, as implemented in the Quantum-ESPRESSO (QE) computational package.³⁴ The correlation and exchange contributions to the total energy were calculated with the Perdew–Wang 91 formulation of the generalized-gradient approximation, claimed to be well suited for surfaces and able to maintain a good compromise between accuracy and computational cost.³⁵ Core–valence electron interactions were described by ultrasoft pseudopotentials, while the Kohn–Sham electronic valence states were expanded in plane-waves up to 50 and 400 Ry (1 Ry $\sim$ 13.61 eV or $\sim$ 313.8 kcal/mol) for kinetic energy and charge density cutoff, respectively. Atomic positions were optimized using the Broyden–Fletcher–Goldfarb–Shanno (BFGS) algorithm as implemented in QE. Spin-polarization was considered throughout all calculations.

Lattice-parameters of bulk MgO were obtained by sampling the Brillouin zone (BZ) in a k -mesh grid of $15 \times 15 \times 15$ Monkhorst–Pack points.³⁶ The values obtained after optimization ($a = b = c = 4.239$ Å) are in good agreement with the experimental data for the B1 rock-salt structure ($a = b = c = 4.217$ Å).³⁷ We used only the Γ -point for sampling the surface BZ of the supercell with a Gaussian smearing of 0.02 Ry for band occupations around the Fermi level. Self-consistency was achieved when the force applied on each atom was less than 1×10^{-3} Ry/bohr and the variation of the total energy between two consecutive iterations was on the order of 1×10^{-4} Ry. Convergence was initially confirmed with respect to the k -point sampling, cutoff energy, and the number of layers, so that the methodology is reasonable and the theoretical results are reliable.

The climbing-image nudged elastic band (CI-NEB) method³⁸ was used to locate energy barriers through the minimum energy path (MEP) by connecting two minima: initial (IS) and final (FS) states. To locate the configurations of maximum energy along the MEP, which climb uphill to the saddle-point and are identified as the transition states (TSs), we specified eight images of elementary steps that span the space between the optimized reactant and product geometries. Then, the direct (forward) activation energy (E_a) was estimated as the energy difference between TS and IS. The motion of surface atoms at high temperatures has not been explicitly included into our calculations, based on the fact that temperature corrections ($k_B T$) are usually small when compared to dissociation barriers at high temperatures and do not significantly influence mechanistic trends.³⁹ Aiming at characterizing the nature of the interaction between EtOH and MgO surfaces, a detailed analysis of the electronic structure was carried out: charge density difference (ρ_{diff}) plots were

obtained by subtracting the sum of the charge densities of the molecule plus the clean surface (in the same geometry) from the charge density of the total adsorbate system, defined as $\rho_{\text{diff}} = \rho - \rho_{\text{EtOH}} - \rho_{\text{MgO}}$, where ρ denotes the charge density of EtOH adsorbed on the MgO surfaces, and ρ_{EtOH} and ρ_{MgO} represent the charge density of the isolated EtOH molecule and of the supercell representing the surfaces, respectively.

2.2. Surface Models. The supercells used to investigate EtOH adsorption and surface reactions were modeled by employing periodic boundary conditions. For that, the DFT-optimized MgO unit cell vectors were periodically translated three times along the *a* and *b* crystallographic axes and twice along *c*, generating a supercell containing 144 atoms divided into four atomic layers and with a surface area of $\sim 1.12 \times 10^{-18} \text{ m}^2$, large enough to avoid molecular interactions between periodic images. Such an approach represents a (100) Miller plane, which stands for the energetically most stable MgO-periclase facet.⁴⁰ In spite of the fact that the catalytic properties of periclase are believed to emerge from a complex interplay between surface Lewis acid (Mg^{2+}) and basic (O^{2-}) properties, the nature and the composition of the active sites are not yet unambiguously established.²¹ Many groups have studied adsorption and reaction mechanisms of several organic molecules on MgO.^{22,41–46} As a common point, it is generally accepted that lower atomic coordination environments for Mg and O are more reactive.^{9,47} Consequently, the catalytic activity would be related to the presence of surface defects claimed to exist in different coordination environments according to their locations, as experimentally evidenced through X-ray and atomic force microscopy investigations.^{48,49}

Surface defects exhibit a wide variety of local environments like 5-fold ($\text{Mg}_{5c}/\text{O}_{5c}$), 4-fold ($\text{Mg}_{4c}/\text{O}_{4c}$), 3-fold ($\text{Mg}_{3c}/\text{O}_{3c}$) coordinated sites and combinations thereof. In order to probe such a chemical environment, a defected system consisting of 123 atoms (Figure 1) has been created by cutting out the

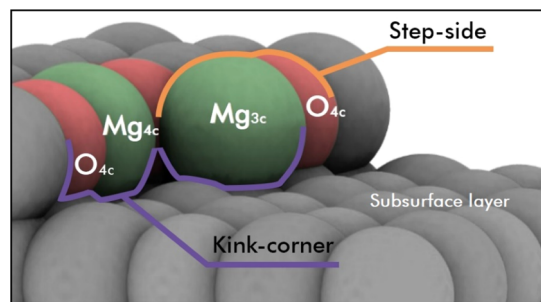


Figure 1. Frame of the supercell used in DFT calculations showing the first two atom-layers. The regions sharing the Mg_{3c} cation, namely kink-corner (purple line) and step-side (orange line) are also highlighted. Henceforth, we use the term active center to denote this ensemble consisting of lower-coordinated Mg cations and O anions. The coordination numbers of Mg (green) and O (salmon) sites are shown.

corresponding three-dimensional full 144-atom supercell. Hence, we were able to evaluate specifically one of the most active types of surface defects, that is, kink-type defects (Figure 1) that consist predominantly of 3- and 4-fold coordinated magnesium ($\text{Mg}_{3c/4c}$) and oxygen (O_{4c}) surface sites, respectively. A similar approach has been successfully used in

several studies on the MgO surface and catalytic properties.^{41,50–53}

Surface relaxation was further considered by allowing the two topmost layers along with the adsorbed molecules to freely relax while the bottom two layers were kept fixed in their optimized bulk positions, mimicking a semi-infinite crystal. A vacuum layer of at least 17 Å was added along the *c* axis to minimize spurious interactions between adjacent replicas. The Bengtsson dipole correction⁵⁴ was applied in order to cancel long-range dipole interactions caused by a charge rearrangement on the surface due to adsorption of EtOH and interactions between the two surfaces of the slab.

2.3. Free Energy Diagrams. A statistical thermodynamics treatment was applied a posteriori to the total-energy calculations (0 K) in order to obtain the Gibbs free energy as a function of temperature (*T*) and pressure (*p*), $G(T, p)$, of each species. Here, the calculation of the vibrational modes is the most computationally demanding task. Thus, only the degrees of freedom related to adsorbed species were considered and the vibrations were obtained at the Γ *q*-point. Low-frequency vibrational modes are usually not accurately calculated by DFT methods, although they might contribute considerably to the vibrational entropy. Therefore, in order to roughly correct the harmonic oscillator model breakdown for low-frequency modes, we approximated vibrational frequencies below 100 to 100 cm^{-1} ($\sim 1/2RT$ at room temperature). The calculations were performed with the PHonon code of Baroni et al.⁵⁵ and $G(T, p)$ was obtained in accordance with the harmonic approximation through the analysis of the vibrational frequencies as obtained via density functional perturbation theory (DFPT). We approximated the pressure to 1 bar. Thus

$$G(T, p) = E_e + E_{\text{zpe}} + E_{\text{vib}} + E_{\text{rot}} + E_{\text{trans}} + pV - T(S_{\text{vib}} + S_{\text{trans}} + S_{\text{rot}}) \quad (1)$$

In which, E_e is the electronic energy as obtained directly from DFT calculations, while zero-point vibrational energy (E_{zpe}) corrections are given by

$$E_{\text{zpe}} = \frac{1}{2} \left(\sum_i h\nu_i \right) \quad (2)$$

Vibrational, translational, and rotational energies of gas-phase molecules are defined, respectively, as

$$E_{\text{trans}} = \frac{3}{2} RT \quad (3)$$

$$E_{\text{rot}} = RT \text{ (linear) or } \frac{3}{2} RT \text{ (non-linear molecule)} \quad (4)$$

$$E_{\text{vib}} = R \sum_i \frac{h\nu_i}{k_B} \left(\frac{1}{2} + \frac{1}{e^{h\nu_i/k_B T} - 1} \right) \quad (5)$$

Vibrational (S_{vib}), translational (S_{trans}), and rotational (S_{rot}) entropies are given by eqs 6–8

$$S_{\text{vib}} = R \sum_i \left(\left(\frac{h\nu_i/k_B T}{e^{h\nu_i/k_B T} - 1} \right) - \ln(1 - e^{-h\nu_i/k_B T}) \right) \quad (6)$$

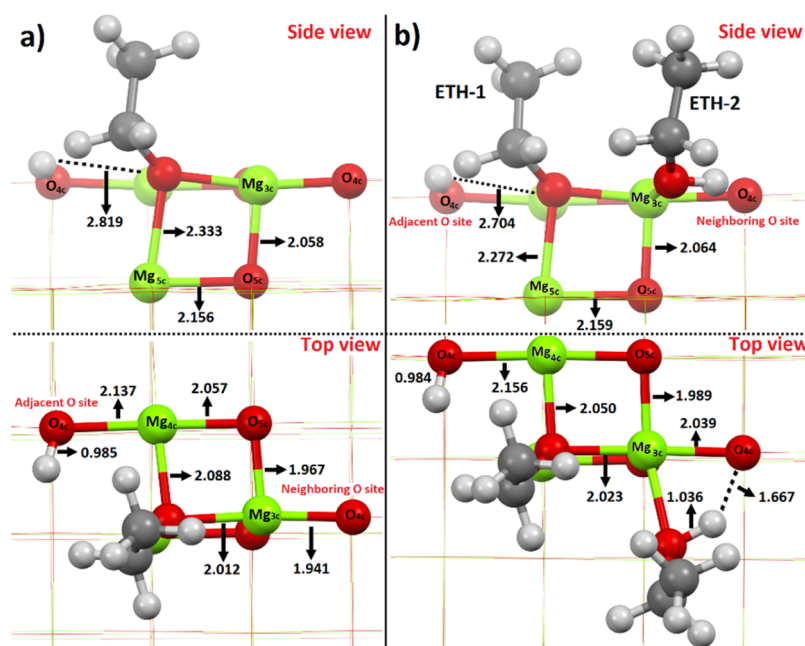


Figure 2. DFT optimized geometries of the EtOH monomer (a) and dimer (b) adsorbed at a kink-defected active-center of MgO. The numbers in bold indicate interatomic bond distances in Å; 3c, 4c, and 5c indicate the coordination number of surface Mg and O sites. 4c O sites (O_{4c}) were divided into adjacent and neighboring (directly bound to Mg_{3c}). Green, red, grey, and white balls represent Mg, O, C, and H atoms, respectively.

$$S_{\text{trans}} = R \ln \left(q_{\text{trans}} + \frac{5}{2} \right), \quad \text{where } q_{\text{trans}} = \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} \frac{k_B T}{P} \quad (7)$$

$$S_{\text{rot}} = R \ln \left(q_{\text{rot}} + \frac{3}{2} \right) \quad (\text{non-linear}),$$

$$\text{where } q_{\text{rot}} = \frac{\sqrt{\pi}}{\sigma} \left(\frac{8\pi^2 k_B T}{h^2} \right) \times \sqrt{I_x \times I_y \times I_z} \quad (8)$$

In eqs 2–9, ν_i , k_B , h , R , m , σ , and I correspond to vibrational frequencies, Boltzmann constant, Planck's constant, universal gas constant, molecular weight, rotational symmetry number, and moments of inertia, respectively. The ideal gas approximation was used to obtain vibrational, translational, and rotational entropic and enthalpic contributions. On the other hand, translational and rotational contributions have been converted into vibrational modes for adsorbed molecules, intermediates, and TSs; the pV work term was neglected because it was considered very small for adsorbed species when compared to the energy-related terms, yielding

$$G(T, p) = E_e + E_{\text{zpe}} + E_{\text{vib}} - TS_{\text{vib}} \quad (9)$$

2.4. Experimental Section. 2.4.1. Catalyst Synthesis and Characterization. In short, our MgO catalyst was prepared via precipitation of $Mg(OH)_2$ by the addition of a solution of KOH and K_2CO_3 to $Mg(NO_3)_2$. $Mg(OH)_2$ was then thermally treated to yield the used MgO. The material was characterized by X-ray fluorescence (XRF), X-ray diffraction (XRD), N_2 physisorption, and CO_2 temperature-programmed desorption (TPD). For full synthesis and characterization descriptions, the reader is kindly referred to our previous work.⁵⁶

2.4.2. Diffuse Reflectance Infrared Fourier Transform Spectroscopy. The tests were conducted in a Bruker VERTEX 70, with a liquid nitrogen-cooled MCT detector, ZnSe windows, and a diffuse reflectance chamber for high-temper-

ature treatments (Harrick). The samples were pretreated at 573 K in 30 mL/min of pure Ar for 1 h and then cooled back to 323 K. Afterward, the catalyst was submitted to a 30 mL/min 1% EtOH in Ar flow at 323 K prior to recording spectra every 5 min for 30 min. Then, the flow was switched back to pure Ar to assess the behavior of EtOH desorption. The spectra were also obtained every 5 min for 30 min after pure Ar started flowing. This procedure was repeated up to 673 K with temperature increments of 50 K.

2.4.3. Bench-Scale Kinetic Experiments. An Agilent 7890A gas chromatograph with an FID detector was used to conduct the experiments. All species were detected using an Agilent CP-PoraPLOT QH column (25 m, 0.32 mm i.d.). Nitrogen (Praxair, ultra-high purity) was used as carrier gas and methane (Praxair, ultra-high purity) was used as internal standard at a 3% concentration. Anhydrous EtOH (Sigma-Aldrich, >99.5%) was fed into the system using a Teledyne ISCO 500D syringe pump. An evaporator kept at 413 K was used to vaporize the incoming EtOH prior to the mix with the carrier gas. A 1/2" tubular Ti reactor was used for all reactions. Internal pressure build-up was monitored using a pressure gauge and the reaction temperature was monitored using a thermocouple inserted into the tube at the catalyst bed height. The lines were kept at 393 K in order to assure that no cold spots were present, thus avoiding condensation and fouling in the tubes. Catalyst masses, N_2 flow rates, and EtOH flow rates were adjusted case-by-case (vide infra) in order to keep contact times constant, system pressure at 1.1 bar, and differential conversions. The catalysts were activated at 773 K under N_2 for 1 h prior to the tests.

Reaction rates for BuOH (r_{BuOH}) are calculated as

$$r_{\text{BuOH}} = \frac{C_{\text{BuOH}} F}{w S_s} \quad (10)$$

where C_{BuOH} is the concentration of BuOH in mol/dm³, measured chromatographically, F is the total volumetric flow

Table 1. Bond Lengths (Å) of the Gas Phase and Adsorbed EtOH on the MgO Surface

		$d(\text{O}-\text{H})$	$d(\text{O}-\text{C}_\alpha)$	$d(\text{C}_\alpha-\text{C}_\beta)$	$d(\text{O}_{\text{hyd}}-\text{Mg})$	$\Delta_{\text{ad}}/\text{eV}/\text{EtOH}^b$
gas phase ^a	monomer	0.974	1.437	1.515		
		0.971 ^a	1.431 ^a	1.512 ^a		
kink ($\text{Mg}_{3c}\text{O}_{4c}$)	monomer	2.818	1.433	1.524	2.008 (Mg_{3c})	−0.97
					2.093 (Mg_{4c})	
	ETH-1	2.731	1.429	1.524	2.008 (Mg_{3c})	−0.46
					2.093 (Mg_{4c})	
	ETH-2	1.035	1.446	1.517	2.128 (Mg_{3c})	

^aExperimental values for EtOH.⁶⁵ ETH-1 or ETH-2 stands for relative position of each EtOH molecule (see Figure 2b). ^bHere, the symbol Δ_{ad} corresponds to adsorption free-energies (in eV/EtOH) estimated for a temperature of 673 K.

rate in cm^3/min , S_g is the catalyst specific surface area in m^2/g , and w is the catalyst mass in the assay in mg. This expression assumes the differential regime, which was experimentally verified [see Supporting Information, Section S2.3].

Ethanol conversion (X_{EtOH}) is given by

$$X_{\text{EtOH}} = \frac{C_{\text{EtOH},0} - C_{\text{EtOH}}}{C_{\text{EtOH},0}} \quad (11)$$

where $C_{\text{EtOH},0}$ is the EtOH concentration measured with the reactor on bypass and C_{EtOH} is the measured EtOH outlet concentration. Blank assays were conducted to verify any undesirable background reactor activity and the EtOH conversion under such circumstances was below 0.4% at 573 K (see Figure S1).

We used a sequential planning of experiments in order to estimate the kinetic parameters in this study. We chose a D-optimal criterion because the main objective of our estimation procedure is the quality of prediction of the model. While it is known that this criterion could lead to inferior parametric quality, it is expected beforehand that kinetic parameters will be intrinsically and unavoidably correlated because of the nature of the Langmuir–Hinshelwood–Hougen–Watson (LHHW) modeling approach.^{8,57,58} A thorough description of this experimental approach, the setup of the parametric estimation problem, and model validation procedures are discussed in Sections S1 and S4.

3. RESULTS

3.1. Adsorption Geometries. Our theoretical predictions indicate that MgO surface terraces, composed of 5-fold coordinated Mg_{5c} and O_{5c} sites, are essentially inert toward EtOH activation (see Section S5.1 for more details), in agreement with the literature.^{41,43,44,59} On the other hand, experimental studies^{60–62} revealed the presence of dissociated EtOH on MgO samples even at temperatures as low as 150 K,⁶³ so that surface terraces give only a minor (if any) contribution to the catalytic activity. Thus, it is important to investigate how EtOH interacts with lower than 5-coordinated surface sites. Our DFT calculations revealed, for instance, that the presence of lower-coordinated Mg_{3c} sites caused a noticeable relaxation at the surroundings. The $\text{O}_{5c}-\text{Mg}_{3c}-\text{O}_{4c}$ angle (Figure 2) in a defective MgO surface is 97.1° , while the $\text{O}_{5c}-\text{Mg}_{5c}-\text{O}_{5c}$ angle in a perfect terrace is 90° . Moreover, the $\text{O}_{4c}-\text{Mg}_{3c}$ and $\text{O}_{5c}-\text{Mg}_{3c}$ bond distances were found to decrease from 2.12 Å in a perfect surface terrace to 1.938 and 1.980 Å, respectively. Thus, it is expected that this local relaxation increases the reactivity upon EtOH. The preferential adsorption geometries of EtOH–monomer ($n = 1$) and EtOH–dimer ($n = 2$) are depicted in Figure 2.

The energetically most stable adsorption configuration of EtOH–monomer correlates with a highly reactive site in the surroundings of the Mg_{3c} site (Figure 2a). The calculated adsorption free energy (Δ_{ad}) at 673 K was -0.97 eV (Table 1). The adsorption process caused spontaneous cleavage of the EtOH hydroxyl (O–H) bond resulting a surface hydroxyl species ($\text{O}_{4c}-\text{H}$). The remaining ethoxy maintains a η_1 (upright) geometry coordinated simultaneously with Mg_{3c} and Mg_{4c} and subsurface (second layer) Mg_{5c} sites. The dissociative adsorption leads to a depletion of charge density (blue isosurface) around the departed H atom confirming the formation of a new $\text{O}_{4c}-\text{H}$ bond (Figure S8a). An accumulation of charge around the $\text{C}_\alpha-\text{H}$ bond changed the length and the $\text{H}-\text{C}_\alpha-\text{H}$ angle from 1.104 to 1.107 Å and from 107.5 to 105.8° , respectively; the $\text{C}_\alpha-\text{C}_\beta$ bond elongated from 1.515 Å (gas) to 1.524 Å. The analysis of charge partitioning carried out by means of the Bader method⁶⁴ shows that the H atom bound to O_{4c} sites ($\text{O}_{4c}-\text{H}$) loses most of its charge being best described as a surface proton (H^+).

Adsorption of an EtOH–dimer (Figure 2b, ETH-2) was found to be less exergonic ($\Delta_{\text{ad}} = -0.47$ eV/EtOH). While the ETH-1 O–H bond breaks spontaneously upon adsorption, the adsorption of a vicinal ETH-2 at the step-side vicinity was nondissociative because its O–H bond was only partially elongated (from 0.974 to 1.035 Å) (we kindly invite the reader to check an animation video in the Supporting Information). This agrees with the fact that molecularly adsorbed EtOH was detected even at high temperatures.²¹ It also suggests that vicinal ETH-2 is positioned on a slightly less-reactive chemical environment (step-side). Further geometrical parameters are shown in Table 1 and Figure 2, and a detailed electronic structure analysis, discussing the different reactivities associated with kink-corner and step-side upon EtOH adsorption, can be found in Section S5.2. The charge-difference plot of ETH-2 (Figure S8b) is consistent with hydrogen (H)-bonding ($\text{O}-\text{H}\cdots\text{O}_{4c}$),^{66,67} which means that the ETH-2 O–H bond might easily be cleaved. A few scenarios were tested, demonstrating that most likely the formation of $\text{O}_{4c}-\text{H}$ occurs (see Table S4—step 1). As a direct consequence, a fast equilibrium between undissociated (O–H) and dissociated ($\text{O}_{4c}-\text{H}$) hydroxyl is expected. In qualitative terms, this agrees with Gines and Iglesia¹⁵ who suggested that, for a K–Ce-promoted MgO catalyst, EtOH O–H bonds might be cleaved and reformed several times while adsorbed intermediates stay on the surface. It is also in consonance with Liang and Gay,⁶⁸ who, by using ^{13}C solid-state nuclear magnetic resonance (SS-NMR), suggested the existence of two different EtOH species on MgO, (i) a more strongly bound ethoxy associated with cross-polarization (long ^{13}C spin–lattice relaxation time) and

(ii) a less strongly bound species associated with excitation pulses at 90°. ⁶⁸

3.2. In Situ DRIFTS at Variable Temperature and Theoretical Calculations. Grounded on the preceding theoretical information, the effect of temperature on the chemical transformations of EtOH was assessed by in situ DRIFTS (Figure 3). Starting at the lowest temperature (323

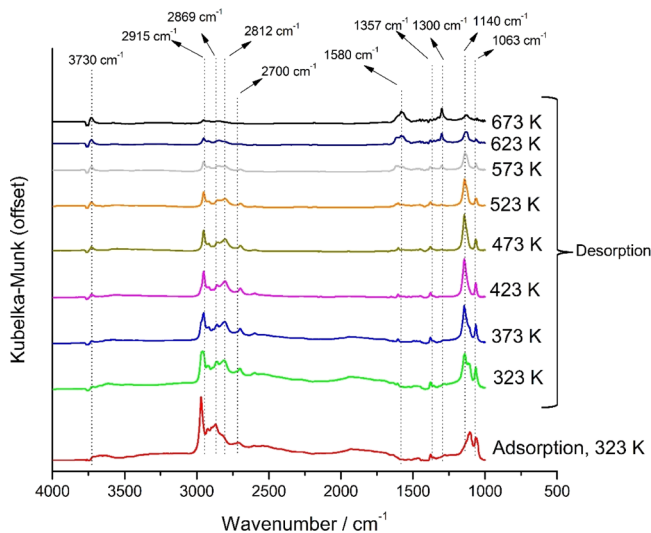


Figure 3. Experimental DRIFTS spectra at the assessed temperatures. Discussed bands are highlighted.

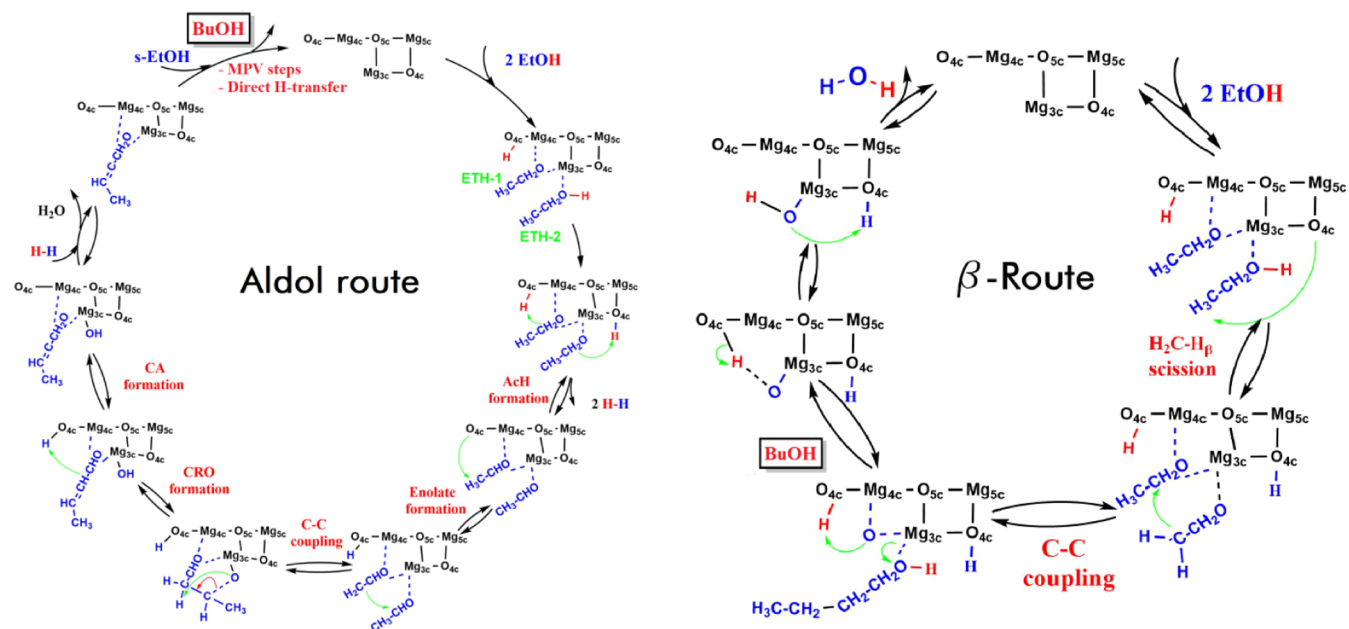
K), the spectra obtained under adsorption and desorption conditions show very broad and weakly intense bands around 3000–3500 cm^{-1} assigned mainly to undissociated EtOH interacting through H-bonds. ^{69,70} Predominant bands around 2700–3000 cm^{-1} were assigned to stretching modes of CH (C–H, C–H₂, and C–H₃), C–C, C–O, and C–C–O bonds that belong to chemi (ethoxy) and physisorbed EtOH, ^{21,22,56} whereas bands at ~ 1360 and ~ 1300 cm^{-1} are likely because of bending and wagging frequencies of C–H₃ and C–H, respectively. On the other hand, while vibrational modes in the ~ 3400 – 3700 cm^{-1} region can be unambiguously assigned to stretching of lower-coordinated hydroxyl ranging from highly basic (isolated) to multicoordinated groups, ^{41,69,71} the assignment of modes in the 1000–1200 and 1200–1600 cm^{-1} intervals are not so obvious. For instance, Birky et al. assigned modes at 1058–1066 cm^{-1} to C–C–O, related to (molecularly) adsorbed EtOH, while the 1119–1132 cm^{-1} region was attributed to ethoxide species coordinated to surface Mg sites. ²¹ In turn, Chierigato and coworkers tentatively assigned the band around 1165 cm^{-1} to a so-called carbanion species generated from the abstraction of a H $_{\beta}$ atom, observing a red-shift to 1124 cm^{-1} as a function of temperature. ²²

Upon increasing temperatures, we observed continuously weaker bands around the 2750–3000 cm^{-1} corroborating a reduction in the concentration of surface ethoxy, in turn indicating that surface species are being progressively decomposed into a plethora of products. ^{20–22,56} Particularly, the assignment of the band around 2700 cm^{-1} is usually difficult because it could be attributed to both C–H stretching of aldehyde methyl groups, surface hydroxyls ⁶⁹ or even assigned to Fermi resonances of ethoxy C–H deformations. ⁷² Therefore, in order to get a better insight into our DRIFTS

experiments, we made use of the DFPT ^{34,55} to obtain vibrational modes of selected theoretical adsorption systems that we believe represent different stages of EtOH-to-BuOH upgrading.

Simulated spectra of adsorbed EtOH and AcH dimers as well as of a new C $_{\beta}$ –H bond-activated intermediate, named OMC-like (see Section 3.3.2), are depicted in Figure S10. The simulated spectrum of the EtOH dimer indicates that the calculated mode at 2685 cm^{-1} corresponds to a frustrated stretching vibration of hydroxyl groups, corroborated by the strong component of the corresponding displacement vector. While on the one hand this agrees with our DFT adsorption system and projected density of states (PDOS) analysis (Section S5.3) confirming the site-dependent half-dissociated nature of EtOH adsorption, ^{15,68,69} on the other hand the calculated band at 2747 cm^{-1} in the simulated spectrum of OMC-like is dominated by the displacement of a surface (O $_{4c}$ –)H atom toward a C $_{\beta}$ (C $_{\beta}$ –H₂ group), resulting in a large amplitude vector. In turn, the simulated spectrum of the AcH dimer (Figure S11) could not reproduce the band around the 2700 cm^{-1} region. Instead, it clearly displays bands related to stretching modes of C–H and C–H₃ bonds in the 2800–3000 cm^{-1} region, whereas bands in the 1300–1500 cm^{-1} region could be assigned to bending modes of the same bonds, ^{69,73} in agreement with our DRIFTS results. Ordonsky et al. ³¹ observed that only in the presence of gas-phase AcH the band around 2700 cm^{-1} represents a fingerprint of the aldehyde CHO group (C–H stretching). Therefore, a blue-shift to the region ~ 2800 cm^{-1} would be expected for aldehydes under thermal desorption conditions as a result of an elongation of the C–O bond (adsorbed phase); ⁶⁹ indeed, our calculations (data not shown) indicate that the C–O bond of adsorbed AcH is longer when compared with the gas-phase AcH. Despite these facts, our DRIFTS experiments point to the presence of a broad low-intense band ~ 2700 cm^{-1} already under adsorption conditions at 323 K. This band persists at higher temperatures losing intensity from 323 K (desorption) until disappearing at 623 K so that the bands assigned to stretching modes of CH groups of aldehydes and other groups are most likely in the 2800–3000 cm^{-1} range, as mentioned above. Thus, besides corroborating the partially dissociated nature of EtOH adsorption, ⁶⁹ the band around 2700 cm^{-1} could also be regarded as a signature mode of the presence of a new adsorbed β -activated intermediate (OMC-like) whose chemistry was thoroughly investigated later in this work.

Regarding lower wavenumber regions, we noticed that the band ~ 1100 cm^{-1} loses intensity and blue shifts to a position ~ 1142 cm^{-1} from 323 K until 473 K. After that point, band intensity starts decreasing until the temperature reaches 673 K. The former band (~ 1100 cm^{-1}) can be attributed to stretching modes of C–O bonds (ethoxy species), while the later one (~ 1142 cm^{-1}) suggests the transformation of ethoxy into C2 and C4 aldehydes and alkoxy species that are formed through a coupling reaction. ^{26,61,69,74} Additionally, a relatively broad band centered around 1600 cm^{-1} appears at 373 K and becomes more intense as temperature increases. Then, it slightly red shifts (~ 1580 cm^{-1}) and broadens from desorption starting at 523 K. This region can be related to the formation of surface systems containing C=C bonds (stretching mode), like CRO and CA. Despite not being able to fit perfectly with the experimental data, the simulated spectra (Figure S12) of systems corresponding to adsorbed CA and CRO clearly indicate the presence of C=C bonds, which are characterized

Scheme 1. Elementary Steps of Aldol and β -Route Reaction Pathways for EtOH Coupling

by relatively large amplitude displacement vectors at 1509 cm^{-1} (CRO) and 1647 cm^{-1} (CA), respectively. Others have also assigned bands in the 1500–1650 cm^{-1} region to the presence of strongly bound carbonyl species as well as residual carbonates and acetates, which also seem to be present at higher temperatures although not captured by our theoretical models.^{22,26,56,61,69,74}

In addition to the previous discussion on lower wavenumber regions, we believe that bands in 1000–1250 cm^{-1} could be somehow masking the presence of hydride (H^-) as a result of the interaction of H_2 (a by-product of AcH formation) with MgO .^{75–77} More specifically, our DRIFTS spectra indicate an emerging band at 1144 cm^{-1} , which falls essentially on the region related to the presence of surface H^- . Experiments suggest that the adsorption of H_2 is stable up to 400 K and from that point the H species could either recombine as H_2 or diffuse along the surface.^{76,78,79} Our DFT calculations indicate a heterolytic interaction of H_2 with kink defects. In this case, the heterolytic dissociation forms a stable hydride (H^-) species bridged to both low-coordinated Mg sites (Mg_{3c} , Mg_{4c}) and to the subsurface Mg_{5c} site, whereas the remaining fragment (H^+) binds to an O_{4c} site forming $\text{O}_{4c}\text{-H}$ (Table S4—step 1). The heterolytic nature of H_2 dissociation can be confirmed by the analysis of charge partitioning, which indicates that $\text{H}^{(-)}$ is negatively charged by 0.82 e^- .^{67,71,75–77,79–81} The calculated H^- - H^+ and O_{4c} -H distances are 2.522 and 0.977 Å, respectively. The simulated vibrations (not shown) indicate that the mode related to Mg-H stretching is 1135 cm^{-1} , while the mode at 3487 cm^{-1} was assigned to O_{4c} -H stretching, in good agreement with previous literature.^{67,75–77} Hence, we assume the existence of surface H^- in one of our proposed reaction pathways, as detailed in Section 3.3.1.5.

3.3. DFT Assessments of C–C Coupling Mechanisms.

Here, we analyzed several surface species identified by DFT and DRIFTS and investigated reaction pathways. Experiments suggest that MgO catalysts are capable of adsorbing large amounts of EtOH, reaching surface coverage (θ) as high as ~50% of the Mg-O pairs; however, only a minor fraction of the activated intermediates (~10%) indeed leads to upgraded

C_4 products.^{20,21} Accordingly, the EtOH-dimer in our DFT cells of simulation corresponds to $\theta \sim 0.11$ monolayer (ML), representing well the experimental chemical environment. It is important to mention that the formation of AcH by means of C_α -H bond scission, followed by aldol condensation, is usually accepted as the main path for C–C coupling over basic catalysts (Guerbet mechanism).^{6,12,20,21} Conversely, others claimed that, as the amount of BuOH did not increase even after the addition of aldol intermediates, this route would be irrelevant to the formation of upgraded C_4 products.^{14,16,29,33} The preference for such a pathway was further reinforced by Chieragato et al.,²² who discarded the presence of C_4 aldehydes or olefins (such as BD), suggesting that butyraldehyde (BuH) and CRO are generated after CA and BuOH.

In order to shed more light on these conflicting issues, we probed the elementary reaction steps dividing them into two main routes, namely, (Section 3.3.1) the aldol route (Guerbet-like) and (Section 3.3.2) the β -route (direct coupling). A series of elementary steps consistent with these routes are illustrated in Scheme 1.

3.3.1. Aldol Route. We assume the formation of two vicinal ethoxy species is followed by the generation of two adsorbed AcH plus two H_2 molecules via C_α -dehydrogenation. For this initial step, only two scenarios are possible in which C_α -H bond-breaking process starts: (i) either at ETH-1 (kink-corner) followed by the same reaction at ETH-2 (step-side) or (ii) vice-versa (C_α -H rupture starting at ETH-2 and then ETH-1). We found that the formation of AcH molecules starting at ETH-1 (i) is kinetically and thermodynamically less favored and, therefore, only the second (ii) reaction channel (i.e., AcH formation starting at ETH-2) will be further considered (aldol route).

3.3.1.1. Acetaldehyde Formation. As shown in Table S4 (steps 2–3) and Figure 4, the forward free-energy barrier to generate AcH first at ETH-2, (ii) $\text{ETH-2} \rightarrow \text{AcH-2} + \text{H}_2$, is 0.99 eV, while the reverse barrier is 0.62 eV, resulting in an endergonic process ($\Delta_r = +0.37$ eV). Next, the formation of the second AcH molecule ($\text{ETH-1} \rightarrow \text{AcH-1} + \text{H}_2$) needs to

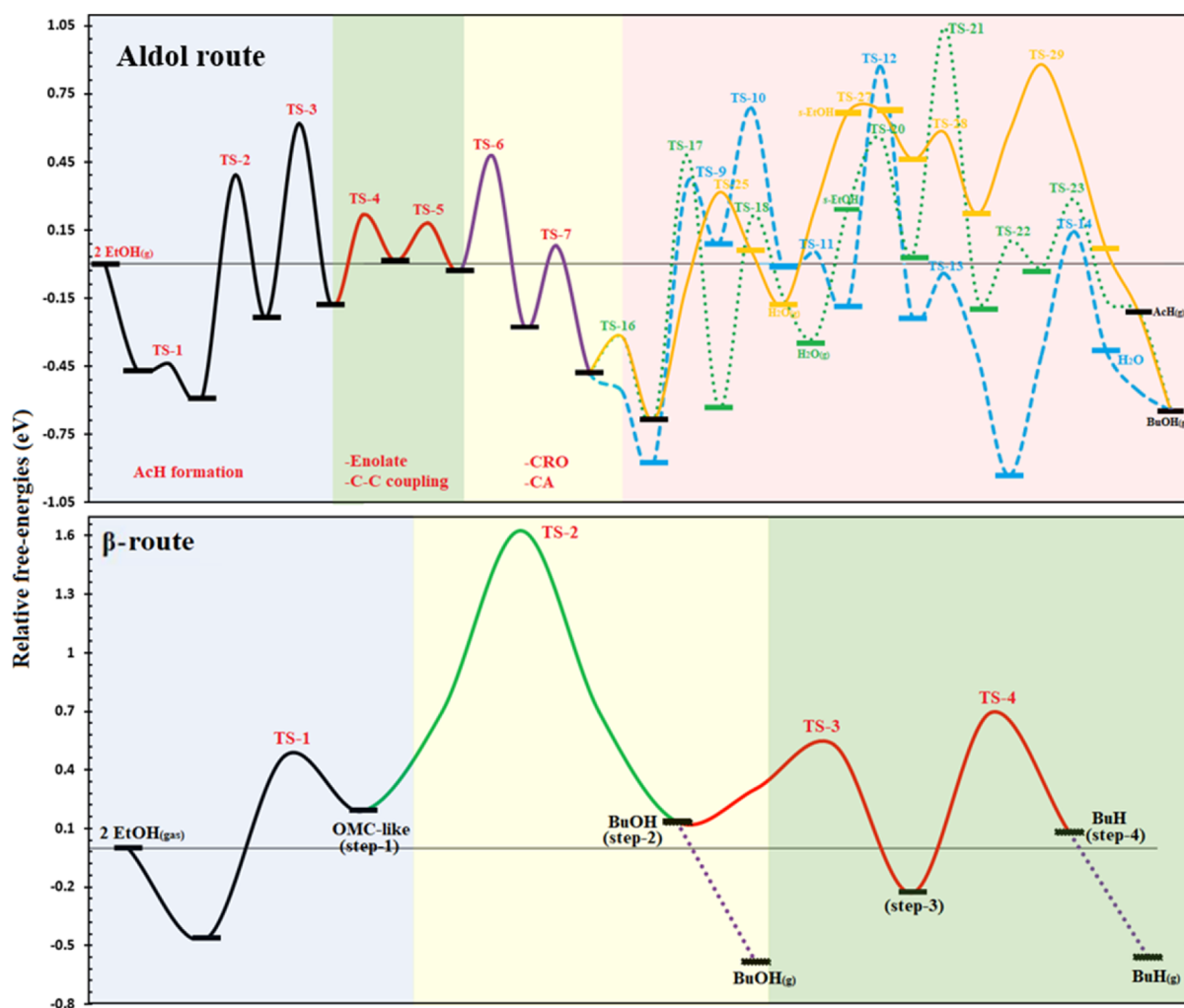


Figure 4. PED for the proposed aldol route (upper panel): green (dotted), orange (full), and blue (dashed) lines correspond to H_2 (Section 3.3.1.5), Meerwein–Ponndorf–Verley (MPV) (Section 3.3.1.6), and the direct H-transfer mechanism (Section 3.3.1.6), respectively; β -route (lower panel) (Section 3.3.2 for more details). The relative Gibbs free energies correspond to a reaction temperature of 673 K. The zero-of-energy was set as two EtOH molecules infinitely separated from the MgO surface. Further details including energetics and snapshots of adsorption geometries and TSs of elementary reactions steps can be found in Section S3.

surpass a lower barrier of 0.85 eV ($\Delta_r = +0.06$ eV), resulting in a lumped activation energy of 1.84 eV. Our results agree with previous literature. Taifan et al.,⁴⁵ using periodic DFT (PBE), found an overall activation barrier of 1.72 eV to form an AcH molecule from adsorbed ethoxide in an endergonic process (+0.25 eV) at 723 K. In a different approach, localized Gaussian basis-set (B3LYP) and atom-frozen clusters were used to determine a relatively higher barrier (~ 1.94 eV) to the formation of AcH.²² On the other hand, Zhang et al.⁴⁶ found lower overall activation energy (1.42 eV) to the formation of AcH on MgO kink defects. We have also probed whether H_α would be prone to be abstracted directly to the subsurface layer (Figure 1), forming $\text{O}_{\text{sc}}\text{--H}$ species. Starting at the ETH-1 position, even after several attempts such a system could not be formed because the H atom migrates back to its initial α -position along the DFT optimization procedure. Concerning ETH-2, we found that H_α might be abstracted, but we were not able to identify a TS for this step. Thus, we believe that initial H_α abstraction by a subsurface O_{sc} site to form AcH is not kinetically determinant to the overall mechanism and, therefore, lower-coordinated (3c and 4c) surface sites are key in the early stages of the reaction.

3.3.1.2. Enolate Formation and C–C Coupling. Enolate could be formed through two different channels starting either at AcH-1 (former ETH-1) or AcH-2 (former ETH-2). Regarding AcH-1, enolate was formed through the reaction $\text{AcH-1} \rightarrow \text{ENO-1} + \text{H}$ (Table S4—step 4) with an activation barrier of 0.39 eV, where H binds to the neighboring $\text{O}_{4\text{c}}$ site resulting in a slightly endergonic final state ($\Delta_r = +0.19$ eV); this barrier is consistent with experimental values found for aldol condensation steps on different catalyst surfaces.^{82,83} On the other hand, the calculated barrier to form enolate from the AcH-2 position through $\text{AcH-2} \rightarrow \text{ENO-2} + \text{H}$ (binding to the adjacent $\text{O}_{4\text{c}}$ site) was found to be considerably higher (1.03 eV) and for that reason this reaction channel (starting at AcH-2) was omitted in this work. Such a distinct chemical behavior derives from the higher basic character of neighboring $\text{O}_{4\text{c}}$ sites, as confirmed by electronic structure analysis (Sections S5.2 and S5.3). The formation of a reactive enolate (ENO-1) starting at AcH-1 was followed by the C–C coupling reaction through a C_β attack (ENO-1) toward the AcH-2 C_α atom forming 3-alkoxybutanal species (Scheme 1; Table S4—step 5). This elementary step is very slightly exergonic ($\Delta_r = -0.03$ eV) and needs to overcome a barrier of only 0.17 eV.

3.3.1.3. CRO Formation. The concerted cleavage of the 3-alkoxybutanal C–O bond and H migration from the C⁽²⁾H₂ group resulted in an adsorbed CRO plus a surface hydroxyl (Table S4—step 6). The calculated energy barrier was moderate (0.51 eV), suggesting that this step is relatively fast, which is consistent with earlier experimental findings;^{11,32,84,85} this process is also thermodynamically favorable (−0.25 eV). Another feasible path would be the hydrogenation of the 3-alkoxybutanal C⁽¹⁾H group generating an adsorbed 3-hydroxybutyrate and then crotyl alkoxide (CA) bypassing CRO (see Table S5). Interestingly, the energetics of the reaction suggests a thermodynamically favorable step ($\Delta_r = -0.64$ eV) with a relatively low activation barrier of 0.44 eV. However, further generation of CA has a barrier of 1.36 eV (compared to 0.51 eV to form CRO), which lets us conclude that CRO is an intermediate in the aldol route and, therefore, CA cannot be generated directly from 3-alkoxybutanal.

3.3.1.4. Crotyl Alkoxide Formation. This step is rather facile because of an activation barrier of 0.36 eV (Table S4—step 7), also being thermodynamically favorable ($\Delta_r = -0.20$ eV), in agreement with thermodynamic studies.^{86–88} The hydrogenating agent (H) was previously generated through the AcH-1 → ENO-1 + H reaction (Table S4—step 4). We carried out additional calculations to check whether this H fragment would be susceptible to migrate far from its initial O_{4c}-bound position along MgO step-edges instead of hydrogenating CRO (see Table S9). We employed a larger supercell and found a diffusion barrier of 1.02 eV ($\Delta_r = -0.37$ eV). This suggests that H prefers to hydrogenate the C-chain rather than hop along step-edges. Despite this fact, we believe that such a barrier is not prohibitively high to fully prevent some of H to eventually diffuse, which could represent an important aspect because in the absence of EtOH (e.g., only AcH in the feed), surface H fragments could be acting as hydrogenating species to generate BuOH.^{12,16,89,90} For instance, in spite of Ndou et al.¹⁶ advocating in favor of a C_β-activated mechanism, the small BuOH yield observed after the passage of pure AcH (with CRO as the main product) could be a consequence of surface-mediated CA hydrogenation. On the other hand, the same authors observed that a reactant mixture of EtOH and AcH decreased (~66%) the relative quantity of CRO by increasing the resulting BuOH yield.¹⁶ Therefore, in the presence of EtOH (pure or mixtures), different mechanisms might coexist.

3.3.1.5. BuOH Formation: H₂ as a Source of H. Here, we assume for a surface-mediated hydrogenation that one H₂ molecule dissociates heterolytically at a kink-corner and that the resulting hydride species (H[−]) can diffuse (Table S6—step 8). Accordingly, we found that it is energetically costly to move H[−] from their stable position with a barrier of 1.23 eV ($\Delta_r = +0.96$ eV) (Table S6—step 9), while the diffusion barrier of protons along the MgO terrace is substantially lower (see Table S10), suggesting they could migrate to locations that are not necessarily the site where H₂ initially dissociated.^{67,75,76,91,92} Regarding the present mechanism, after departing from its initial position the H[−] fragment reaches a subsurface O_{5c} site nearby the CA (Table S6—step 9). From this point, we tested some alternatives and the energetically most favorable pathway proceeds as follows: (i) the hydrogen initially bound to O_{4c} connects with the CA C⁽²⁾ carbon after overcoming a barrier of 0.61 eV with modest exergonicity ($\Delta_r = -0.10$ eV) (Table S6—step 10); (ii) the subsurface O_{5c}-bound H migrates to the O_{4c} site with a small barrier (0.07 eV) ($\Delta_r = -0.17$ eV) (Table S6—step 11); this step is immediately

followed by an intramolecular 1,3 (C(1) → C(3)) H-migration after surpassing a relatively high energy barrier of 1.06 eV ($\Delta_r = -0.05$ eV) (Table S6—step 12); (iv) finally, C⁽¹⁾ carbon is hydrogenated to form adsorbed butoxy after a small barrier of 0.20 eV, resulting in a thermodynamically highly stable system ($\Delta_r = -0.69$ eV) (Table S6—step 13); (v) H₂O is formed through the interaction of the remaining H₂ with a surface hydroxyl (Table S5—step 14). In this case, we considered that H₂ dissociates forming two protons with a barrier of 1.07 eV ($\Delta_r = +0.55$ eV). A H₂O molecule is eliminated and, finally, the removal of BuOH closes the catalytic cycle.

3.3.1.6. BuOH Formation: Hydrogenation via MPV Reaction. The MPV reaction uses an extra molecule, known as a sacrificial alcohol, as the external source of H in order to reduce carbonyl (C=O) groups.^{12,32,90,93} In our approach to this mechanism, we assume that CA, initially in an upright (η_1) geometry, tilts toward the surface assuming a nearly flat configuration with a small barrier of 0.16 eV ($\Delta_r = -0.21$ eV) suggesting that this configuration is favorable (Table S7—step 16). Next, an isomerization reaction occurs, in which a H atom from C⁽¹⁾ migrates to C⁽³⁾ (C⁽¹⁾ → C⁽³⁾) after overcoming a relatively high barrier of 1.17 eV ($\Delta_r = +0.06$ eV) (Table S7—step 17). Thereafter, a H₂ molecule dissociatively adsorbs on the step-side generating two surface H species and one of them reacts with the OH group forming H₂O. This molecule is removed leaving a vacant site where a sacrificial EtOH (s-EtOH) adsorbs ($\Delta_r = +0.40$ eV) (Table S7—step 19). Then, the first H-transfer from s-EtOH to C=O (1st MPV-step) occurs with a relatively low activation barrier of 0.31 eV ($\Delta_r = -0.21$ eV) (Table S7—step 20). Next, a second isomerization takes place through migration of H from C⁽¹⁾ to C⁽²⁾ forming BuH after the system overcomes a barrier of 1.00 eV ($\Delta_r = -0.19$ eV) (Table S7—step 21). Subsequently, the C⁽²⁾–C⁽³⁾ bond rotates with after a barrier of 0.24 eV ($\Delta_r = +0.14$ eV). Then, the second H transfer from the s-EtOH molecule to the BuH carbonyl carbon (2nd MPV-step) happens with a barrier of 0.29 eV ($\Delta_r = -0.15$ eV) generating butoxy species and eliminating an AcH molecule. The corresponding potential energy diagram (PED) is depicted in Figure 4. Finally, BuOH desorbs giving back the clean catalyst surface.

3.3.1.7. BuOH Formation: Hydrogenation via Direct H-Transfer from s-EtOH. In this section, we considered an alternative mechanism in which the CA C=C bond is hydrogenated directly from the s-EtOH molecule. The process starts with CA in a nearly-flat geometry and is followed by H₂ dissociation and H₂O removal, giving place to the adsorption of a s-EtOH molecule. Next, the nearly flat CA molecule rotates ~84° toward s-EtOH almost barrierlessly (0.02 eV); this step is important because it makes the C=C bond more likely to be hydrogenated by the s-EtOH molecule (Table S8—step 25). The process was then divided into two steps: (i) in the first, a H departs from the s-EtOH OH group toward the C⁽²⁾ carbon after a barrier of 0.12 eV ($\Delta_r = -0.24$ eV) (Table S8—step 26); (ii) a butoxy species is formed right away through the transfer of a H from the s-EtOH C_α–H bond to the CA C⁽³⁾ carbon with a reasonable activation energy of 0.66 eV ($\Delta_r = -0.16$ eV) (Table S8—step 27). At the end of this process, an AcH molecule is eliminated. The last step proceeds through BuOH desorption closing the catalytic cycle.

As can be seen in the PED graph (Figure 4), reaction free energies and activation barriers are overall higher regarding the MPV pathway (Section 3.3.1.6) compared to the direct hydrogenation mechanism. Furthermore, we noticed that the

CA isomerization reaction (H-migration from C⁽¹⁾ to C⁽³⁾) has a relatively high barrier (1.17 eV), considering the aldol route rate-limiting step as the formation of two vicinal AcH molecules. Thus, our theoretical models led us to infer that the hydrogenation of CA to BuOH occurs preferentially by means of the direct H-transfer mechanism throughout the aldol route. Nevertheless, it is important to stress that this proposal represents an alternative path based on our theoretical models and, therefore, it does not exclude the occurrence of classic MPV steps. In fact, Young and Davis⁹⁴ claimed that the C=C bond (ethene) could not be hydrogenated prior to the C=O bond (acetone) over MgO at 473 K because a sequence of isomerization steps would be needed before the final carbonyl carbon hydrogenation via MPV steps. Conversely, Moteki and Flaherty⁹⁰ argued that the hydrogenation of C=C bonds of crotyl alcohol occurs by surface hydrogenation. Thus, there is still room for discussion in the future so that further theoretical models need to be considered in order to fully describe the considerable number of possible mechanistic steps.

3.3.2. Direct Coupling: β -Route. We now examine a different situation in which the leading mechanism for EtOH upgrading was not based on the C α -activated steps (aldol route). Instead, C–C coupling initiates through the activation of a C β –H bond (EtOH methyl group) generating a key-intermediate, which in turn reacts with a vicinal ethoxide yielding BuOH as a primary product. For that, we assumed that the adjacent O $_{4c}$ site is capable of activating a H $_{\beta}$ (ETH-2) generating a surface hydroxyl (O $_{4c}$ –H). Within our approach, a H $_{\beta}$ could only be abstracted from ETH-2, so that DFT structure-optimization attempts regarding ETH-1 always bring H $_{\beta}$ back to their initial (ethoxy) position. Thus, we considered as the starting point of the β -route an adsorbed system containing ethoxy (ETH-1) and its vicinal undissociated ETH-2 moiety.

The intermediate has a cyclic C β -activated geometry in which the deprotonated C β carbon is stabilized by the interaction with the surface Mg $_{3c}$ site (see Figure S9). Interestingly, the geometry is similar to a bidentate oxametallacyclic (OMC) species identified in various EtOH-adsorbing systems and, therefore, we named it oxametallacycle-like (OMC-like) intermediate. Idriss et al. proposed the existence of a similar five-membered (–C $_{\beta}$ H $_2$ C $_{\alpha}$ H $_2$ O*) ring formed from β -cleavage on Ru/CeO $_2$ ⁹⁵ and Ru/CeO $_2$ –ZrO $_2$ ⁹⁶ catalysts, while on Rh(111) surfaces it was suggested that OMC intermediates may form ethane and even facilitate the scission of C $_{\alpha}$ –C β bonds.⁹⁷ Chieragato et al.²² also advocated on the existence of a carbanion species generated from H $_{\beta}$ abstraction. The activation energy to cleave the ETH-2 C β –H bond was found to be 0.93 eV and considerably endergonic ($\Delta_r = +0.65$ eV) (Table S11—step 1).

3.3.2.1. C–C Coupling Reaction. BuOH was generated through direct coupling of OMC-like and the vicinal ETH-1 (Table S11—step 2). The calculated barrier was 1.43 eV and the final system is energetically stable ($\Delta_r = -0.06$ eV). We further investigated whether sequential intramolecular dehydrogenations might generate kinetically consecutive products, that is, BuH and CRO, both detected in experiments at 673 K.⁵⁶ For that, the BuOH hydroxyl group first loses its proton after an activation barrier of 0.40 eV ($\Delta_r = -0.36$ eV) (Table S11—step 3). Then, a H $_{\alpha}$ departs to generate O $_{4c}$ –H plus BuH after an activation barrier of 0.91 eV ($\Delta_r = +0.31$ eV) (Table S11—step 4). Finally, we examined whether an

intramolecular dehydrogenation of BuH might generate CRO. However, we could not identify a TS for such an elementary step and hence we discarded this possibility. Thus, BuOH can be formed as a kinetically primary product via a β -route and then BuH as a consecutive product of BuOH. On the other hand, our theoretical data do not support the formation of CRO as a product of sequential dehydrogenation steps (BuOH $\rightarrow$ BuH $\rightarrow$ CRO). In order to test this hypothesis experimentally, we performed a reverse conversion catalytic measurement, in which EtOH was replaced by BuOH in the reactant feed (see Section S2.5). The test confirmed that, even at 673 K, CRO could not be detected, with BuH as the main product.

As we observed, the interaction of ETH-2 with the step-side portion of the active center (Figure 2) does not cause the spontaneous rupture of the O–H bond (necessary condition to form the OMC-like intermediate) because of a lower basicity of the O $_{4c}$ adjacent site (directly attached to Mg $_{3c}$). Despite this fact, the activation barrier to cleave the ETH-2 O–H (Table S4—step 1) bond is quite low (0.04 eV; $\Delta_r = -0.14$ eV) suggesting thereby an equilibrium that is rather sensitive to changes in temperature.¹⁵ In other words, increasing temperatures would have the tendency to hamper the formation of an OMC-like intermediate. Experiments show BuOH (traces) over MgO at $\sim$ 523 K, although other C4 products like CRO and BD are absent;²² the authors claimed that the EtOH-to-BuOH upgrading is fully governed by a carbanion chemistry and that AcH and CRO are not intermediates at all.²² However, such a mechanism becomes inadequate to explain the presence of certain reaction products like CRO and BD upon increasing the reaction temperature to levels higher than $\sim$ 523 K.^{8,20,21,25,98,99}

3.4. Mechanistic Proposal and Kinetic Experiments.

So far, our calculations and experiments (Section S2.5) agree that consecutive generation of BuH from BuOH (BuOH $\rightarrow$ BuH) is feasible, although the sequential formation of CRO is unlikely. In turn, this agrees with experimental arguments suggesting that BuOH and BuH can be generated through a C β -activated pathway. However, increasing temperatures makes the activation of EtOH C β –H bonds unfavorable and, therefore, the surface concentration of aldol intermediates tends to increase through the activation of C $_{\alpha}$ –H bonds, thus favoring the more endergonic aldol route. Based on these findings, we propose that the formation of BuOH is dominated by the β -route pathway under lower reaction temperature conditions, until $\sim$ 573 K (see Table S2), and, from that point on, more aldol products will be formed altering the C–C coupling mechanism to a dominance of the aldol route.

Note that the band intensity in the $\sim$ 2700 cm $^{-1}$ region (Section 3.2), assigned as a signature mode of the β -activated intermediate, becomes very low at 573 K being essentially nonexistent at higher temperatures (623 and 673 K). This adds support to our assumption, although it is difficult (or even impossible) to determine a turning point because of the several variables interfering under reaction conditions (e.g., temperature gradients, nature of the catalyst surface, etc.). Therefore, it is reasonable that both mechanisms coexist. In fact, the work of Ndou et al.¹⁶ confirmed that BuOH yields are very low until 573 K and increase substantially when the reaction temperature rises above 623 K, although they were not assertive regarding a mechanistic proposal suggesting a fully dominating C β -activated pathway.

Table 2. Bench-Scale Experimental Results at 673 K and 1.1 bar Total Pressure

experiment number	catalyst mass/mg	ethanol partial pressure/kPa	helium flow rate/cm ³ ·s ⁻¹	r_{BuOH}^a /nmol·s ⁻¹ ·m ⁻²	ethanol conversion/%	carbon balance/%
1	100	5	26.2	1.37	9	89
2	100	10	27.7	1.44	6	88
3	100	15	28.8	1.47	5	96
4	160	3	41.2	0.782	7	90
5	100	18	29.6	1.58	3	96
6	100	12.5	28.1	1.79	5	91
7	100	20	30	1.94	3	92

^aAverage rate of BuOH formation over three experiments.

In order to get some more experimental evidence, we carried out further catalytic activity measurements at 573 K (Section S2.4). This temperature was chosen because it is nearly in between those previously evaluated by us and others (673 K)^{8,21,56} and by Chieregato et al. (523 K).²² In addition to high yields of AcH (by far the most abundant product), we were able to identify both BuH and BuOH, but no CRO was detected at all (Table S2); when the catalytic measurement was carried out at 673 K, we could indeed identify the presence of CRO.⁵⁶ Therefore, we can conclude that increasing temperatures will inevitably lead to the formation of more aldol intermediates and consequently more CRO and then BuOH will be generated. Because only trace yields of C4 products were detected at temperatures lower than ~573 K, meaningful amounts of BuOH could be only generated through the Guerbet-like aldol route under higher temperature conditions.

In order to further delve into this mechanism, we conducted kinetic experiments aiming at estimating accurate reaction parameters, defining an appropriate mathematical model, and verifying our computational predictions. Our bench-scale kinetic experiments were carried out at 673 K and the reaction mechanism used in the kinetic modeling scheme was based strictly on the elementary steps contained in the aldol route, as proposed by our DFT calculations; further details can be found in Sections S1–S4. The final rate expression obtained for the formation of BuOH from EtOH was

$$r_{\text{BuOH}} = \frac{P_{\text{EtOH}}^2}{\alpha P_{\text{EtOH}}^2 + \beta} \quad (12)$$

where P_{EtOH} is the EtOH partial pressure in the feed and α and β are combinations of the intrinsic rates and equilibrium constants of several elementary steps. The results of the kinetic tests conducted with our sequential experimental planning are summarized in Table 2. Differential reactor behavior and catalyst stability were checked and these results are displayed in Sections S2.2 and S2.3.

Experiments 1–3 refer to the initial set described in Section S1, whose conditions were taken based upon previous literature.^{20–23} Experiments 4–6 are the ones planned computationally with the sequential approach. Our mesh was designed to go from 0.5 to 20% EtOH partial pressure in 0.5% steps, in order to be adequate to instrumental accuracy limitations. Experiment 7 was inserted to check the quality of the model against slight extrapolations, that is, it was not used to estimate the parameters of α and β . The final parametric estimates are shown in Table 3. Thorough model validation results and procedures are shown in Sections S1 and S4, respectively.

Table 3. Estimated Kinetic Parameters

parameter	$\alpha/\text{m}^2\cdot\text{s}\cdot\text{nmol}^{-1}$	$\beta/\text{m}^2\cdot\text{s}\cdot\text{nmol}^{-1}\cdot\text{kPa}^2$
value	0.55 ± 0.04	9.0 ± 0.3

Figure 5 depicts the experimental data along with the model regarding the kinetics of the formation of BuOH. The model

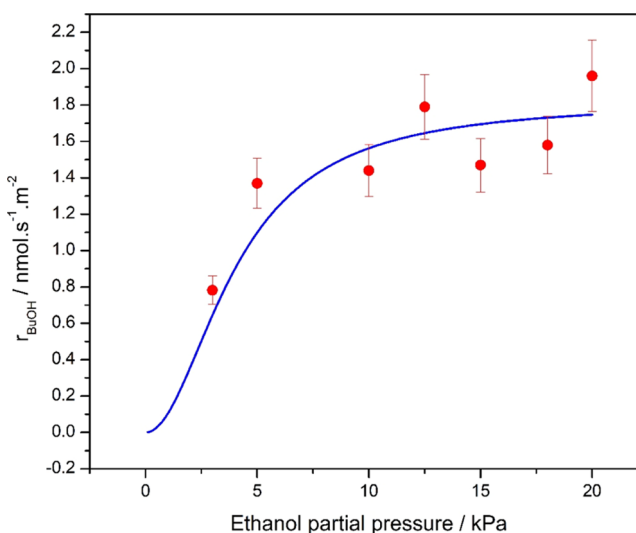


Figure 5. Experimental data (●) and model fit (—) based on the aldol route pathway to form BuOH at 673 K and 1.1 bar total pressure.

seems well suited for use within an EtOH partial pressure range of 0–20 kPa at 673 K. Even with the simplifying hypotheses, the model described the experimental points satisfactorily.

4. CONCLUDING REMARKS

Distinct mechanistic pathways of the EtOH-to-BuOH upgrading reaction on a MgO substrate were thoroughly examined using a combination of DFT calculations, DRIFTS, and catalytic measurements. First, we demonstrated that an appropriate combination of reactive surface sites with the correct alignment of adsorbed EtOH is a necessary condition to promote C–C coupling over MgO. On the other hand, it was shown that O_{sc} and Mg_{sc} sites, which correspond the larger fraction of surface sites (terraces),^{20,21} are not kinetically determinant to upgrading reactions. The catalytic active center was considered as an arrangement of lower-coordinated $\text{Mg}_{4\text{c}}$ and $\text{Mg}_{3\text{c}}$ with neighboring and adjacent (connected to $\text{Mg}_{3\text{c}}$) $\text{O}_{4\text{c}}$ sites and represented by kink-corners and step-sides (Figure 1). Our surface model is consistent with estimates of

the relative concentration of surface defects ($\sim 5\text{--}10\%$) suitable at activating EtOH.^{25,71} In that sense, vicinal, namely, step-side adjacent (less basic) and kink-corner neighboring (more basic), surface O_{4c} sites interact in different ways with EtOH, as demonstrated by PDOS and adsorption studies. Under lower surface coverages ($\theta \sim 0.05$), reactive kink-corners tend to be populated first because of their larger reactivity. As coverage increases ($\theta \sim 0.10$), it becomes more likely for vicinal EtOH molecules (i.e., ETH-1 and ETH-2) to align and thus to undergo coupling reactions. The first mechanistic proposal (aldol route) was based on the EtOH C_α -activated Guerbet-like chain of reactions,²⁰ whereas the second one was based on a direct C_β -activated²⁹ C–C coupling.

Regarding the so-called aldol route, the formation of adjoining AcH molecules (η_1 geometry) at the active center represents the overall rate-controlling step (Figure 2b). The formation of enolate and then C–C coupling to form 3-alkoxybutanal proceed relatively easily, as well as the formation of an adsorbed CRO. Then, CRO undergoes further hydrogenation to generate CA. These steps are common to the aldol route, but the source of hydrogenating species that leads ultimately to BuOH was hitherto not clear. Among the feasible routes probed, direct reduction through the MPV reaction as well as surface-mediated hydrogenation by surface H species have been analyzed more carefully. We proposed a slightly modified version of the MPV reaction between CA and s-EtOH as the main path for BuOH formation.

Concerning the β -route pathway, the fact that ETH-2 adsorption at the step-side does not cause the rupture of the hydroxyl bond is key to the formation of an OMC-like intermediate. In other words, the adjacent O_{4c} site is also capable of activating an ETH-2 C_β –H bond instead of forming exclusively ethoxide. Nevertheless, the low forward energy barrier to cleave the (ETH-2) O–H bond indicates a temperature-dependent equilibrium, which hampers the formation of an OMC-like intermediate at higher temperatures. In turn, higher temperatures increase the surface concentration of aldol intermediates favoring the aldol route pathway. Thus, both the β -route and aldol route pathways would probably exist in a certain temperature range.

It is important to emphasize that our project was able to successfully correlate the fundamental calculations and spectroscopic results with the bench-scale data, something seldom seen in previous works on the subject. On the one hand, works focusing on theoretical results usually do not delve into kinetic modeling to validate their findings;^{22,48,75} on the other hand, studies with a more experimental leaning normally lack the computational approach.^{6–8,11,14,20,21} Using a hybrid theoretical and computation strategy, coupled to an accurate parametric estimation setup, our results highlight an adequate confluence of DFT-derived information, DRIFTS data, and bench-scale kinetic results as a way to derive a fundamentally solid mathematical expression for the kinetic behavior of the system. The validated rate expression provides an adequate benchmark for catalytic improvements and for process upscaling as a way to contribute to making the industrial upgrading of EtOH to BuOH economically viable.

With the help of computational resources, it is possible to choose desirable characteristics of a particular catalyst, such as, in our case, the presence of specific defective sites. This reiterates the need of a more intimate dialogue between computational and experimental works, as to pave the way for the rational development of materials. Therefore, we propose

that synthesis procedures that favor the formation of surface defects and increase the number of lower-coordinated sites may be attractive for the EtOH-to-BuOH upgrading. In the specific case of MgO catalysts, a series of work by Bailly et al. suggests that a sol–gel preparation approach is a potential candidate to achieve this goal^{100–102} and thus to improve MgO's activity in this conversion.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acscatal.0c04616>.

Animation of the adsorption pattern of ethanol on the proposed active center of MgO (AVI)

Kinetic modeling procedure and model validation, additional bench-scale tests, illustration of EtOH coupling mechanisms, methodology and statistical approach for parametric estimation and experimental planning, further theoretical results, references, and atomic coordinates (xyz) (PDF)

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Notes

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■ ABBREVIATIONS

AcH	acetaldehyde
BD	1,3-butadiene
BFGS	Broyden–Fletcher–Goldfarb–Shano
BuH	butyraldehyde
BuOH	1-butanol
BZ	Brillouin zone
CA	crotyl alkoxide
CI-NEB	climbing-image nudged elastic band
CRO	crotonaldehyde
DFPT	density functional perturbation theory
DFT	density functional theory
DRIFTS	diffuse reflectance infrared Fourier transform spectroscopy
ENO	reactive enolate
ETH-1	ethanol adsorbed on an adjacent O site
ETH-2	ethanol adsorbed on a neighboring O site
EtOH	ethanol
FS	final state
IS	initial state
LHHW	Langmuir–Hinshelwood–Hougen–Watson
MEP	minimum energy path
ML	monolayer
MPV	Meerwein–Ponndorf–Verley
OMC	oxametallacycle
PDOS	projected density of states
PED	potential energy diagram
QE	quantum ESPRESSO
s-EtOH	sacrificial ethanol
SS-NMR	solid-state nuclear magnetic resonance
TPD	temperature-programmed desorption
TS	transition state
XRD	X-ray diffraction
XRF	X-ray fluorescence

Nomenclature

C_{BuOH}	1-butanol concentration
C_{EtOH}	ethanol concentration at the reactor outlet
$C_{\text{EtOH},0}$	ethanol concentration on a bypass configuration
$d(a-b)$	bond length between atoms a and b
E_a	activation energy
E_e	electronic energy
E_{rot}	rotational energy
E_{trans}	translational energy
E_{vib}	vibrational energy
E_{zpe}	zero-point energy
F	volumetric flow rate
$G(T,p)$	Gibbs free energy
h	Planck constant
I	moment of inertia
k_b	Boltzmann constant
M	molecular weight
p	absolute pressure
P_{EtOH}	ethanol partial pressure
R	universal gas constant
r_{BuOH}	rate of 1-butanol formation
S_{rot}	rotational entropy
S_{vib}	vibrational entropy
S_{trans}	translational entropy
T	temperature
w	catalyst bed mass
X_{EtOH}	ethanol conversion
α, β	rate expression parameters

Δ_{ad}	ethanol adsorption energy
Δ_r	energy of reaction
θ	surface coverage
ρ	charge density of ethanol on MgO surfaces
ν	vibrational frequency
ρ_{diff}	charge density difference
ρ_{EtOH}	charge density of isolated ethanol
ρ_{MgO}	charge density of isolated MgO surface
σ	rotational symmetry number

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